

AHOP Deployment Guide

1. Local development (XAMPP / Windows)

Requirements

- PHP 8.2+
- Composer
- MySQL (MariaDB)
- Apache with `mod_rewrite`

Steps

```
cd C:\PUP\htdocs\snipe-it
composer install
copy .env.example .env
php artisan key:generate
```

Edit `.env`:

```
APP_URL=http://localhost/snipe-it/public
DB_DATABASE=snipeit
DB_USERNAME=root
DB_PASSWORD=
AHOP_CLINICAL_SIDEBAR=true
AHOP_CLINICAL_DATABASE_ENABLED=false
```

```
php artisan migrate --force
php artisan migrate --path=database/migrations/clinical --force
php artisan ahop:seed-all --password=demo1234
```

Open: `http://localhost/snipe-it/public`

2. Google Cloud Platform (production demo)

Infrastructure

Component	Detail
Provider	Google Cloud Platform
VM	Ubuntu 24.04 LTS

Component	Detail
Web server	Apache 2.4
Database	MariaDB (MySQL compatible)
PHP	8.3
Public URL	https://ahop.jhewhome.xyz
DNS	Cloudflare A record ahop → GCP VM IP
TLS	Let's Encrypt (certbot --apache)

Server stack install

```
sudo apt update
sudo apt install -y apache2 mariadb-server php8.3-fpm php8.3-mysql php8.3-xml \
  php8.3-mbstring php8.3-curl php8.3-gd php8.3-zip php8.3-bcmath php8.3-intl \
  composer git unzip
```

Database

```
sudo mysql -e "CREATE DATABASE snipeit; CREATE USER 'snipeit'@'localhost'
IDENTIFIED BY 'PASSWORD'; GRANT ALL ON snipeit.* TO 'snipeit'@'localhost'; FLUSH
PRIVILEGES;"
```

Clone from GitHub

```
cd /var/www/html
sudo git clone https://github.com/jhewhome/ahop-snipe-it.git snipe-it
cd snipe-it
cp .env.example .env
nano .env # set APP_URL=https://ahop.jhewhome.xyz, DB_*, AHOP_* flags
composer install --no-dev --optimize-autoloader --no-scripts
```

Apache virtual host

DocumentRoot must be **/var/www/html/snipe-it/public**

Example **/etc/apache2/sites-available/ahop.jhewhome.xyz.conf**:

```
<VirtualHost *:80>
  ServerName ahop.jhewhome.xyz
  DocumentRoot /var/www/html/snipe-it/public
  <Directory /var/www/html/snipe-it/public>
```

```
    AllowOverride All
    Require all granted
</Directory>
</VirtualHost>
```

```
sudo a2enmod rewrite
sudo a2ensite ahop.jhewhome.xyz.conf
sudo systemctl reload apache2
```

Custom domain and HTTPS

1. **Cloudflare** (domain registered at Namecheap): add site, point Namecheap nameservers to Cloudflare, add **A** record **ahop** → GCP VM IP.
2. **Let's Encrypt** on the VM:

```
sudo apt install -y certbot python3-certbot-apache
sudo certbot --apache -d ahop.jhewhome.xyz
```

3. Update **.env** after SSL:

```
APP_URL=https://ahop.jhewhome.xyz
APP_FORCE_TLS=true
SECURE_COOKIES=true
```

```
cd /var/www/html/snipe-it
php artisan config:clear
php artisan config:cache
```

Deploy commands

```
php artisan key:generate
php artisan migrate --force
php artisan migrate --path=database/migrations/clinical --force
php artisan ahop:seed-all --password=demo1234
php artisan config:cache
sudo chown -R www-data:www-data storage bootstrap/cache
```

Scheduler (backups & appointment reminders)

```
sudo crontab -u www-data -e
```

```
* * * * * cd /var/www/html/snipe-it && php artisan schedule:run >> /dev/null 2>&1
```

3. Git sync workflow (PC → GitHub → Cloud)

On development PC

```
cd C:\PUP\htdocs\snipe-it
git add .
git commit -m "Describe change"
git push origin main
```

On Google Cloud VM

```
~/deploy-ahop.sh
```

Example `~/deploy-ahop.sh`:

```
#!/bin/bash
set -e
cd /var/www/html/snipe-it
git pull origin main
composer install --no-dev --optimize-autoloader --no-scripts
php artisan migrate --force
php artisan migrate --path=database/migrations/clinical --force
php artisan config:cache
php artisan route:cache
php artisan view:cache
sudo chown -R www-data:www-data storage bootstrap/cache
echo "Deploy complete."
```

4. Environment variables (AHOP)

Variable	Recommended	Purpose
AHOP_CLINICAL_SIDEBAR	true	Clinical menu (patients, OPD, lab, billing)
AHOP_THEME_ENABLED	true	Teal AgilityCare theme

Variable	Recommended	Purpose
AHOP_CLINICAL_DASHBOARD	true	Operations dashboard
AHOP_CLINICAL_DATABASE_ENABLED	false	Single MySQL (local/cloud demo)
AHOP_DAILY_BACKUP	true	Scheduled backups
APP_URL	Full URL with https:// (production)	Required — must match browser URL
APP_FORCE_TLS	true (with HTTPS)	Force HTTPS redirects
SECURE_COOKIES	true (with HTTPS)	Secure session cookies

5. Troubleshooting

Problem	Solution
403 GitHub clone	Use Personal Access Token with <code>repo</code> scope
<code>vendor/</code> missing	<code>composer install</code> with write permissions
Redirect loop	Match <code>APP_URL</code> scheme to how you open the site; align <code>APP_FORCE_TLS</code> / <code>SECURE_COOKIES</code>
Could not open input file: <code>artisan</code>	Run commands from <code>/var/www/html/snipe-it</code>
Certbot Apache plugin missing	<code>sudo apt install -y python3-certbot-apache</code>
Missing <code>patients.email</code>	<code>ALTER TABLE patients ADD COLUMN email VARCHAR(150) NULL;</code>
Permission denied SFTP	Upload to home folder, <code>sudo rsync</code> to web root